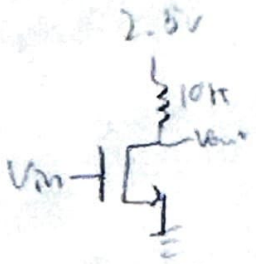
Pylon Training

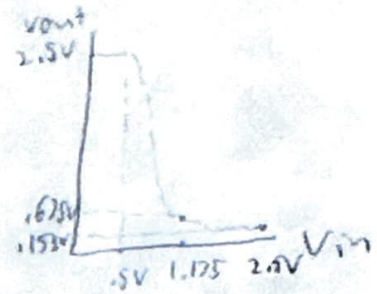
1.



$V_T = .5$

$k_n = 40uA$

b.



$$\frac{V_{DD} - V_{DS}}{10k} = 40u\frac{1}{2}(2.0)(V_{DS})^2$$

$$\frac{2.5 - V_{DS}}{10k} = 40uA\left(\frac{1}{2}\right)((2.5-.5)V_{DS} - \frac{V_{DS}^2}{2})$$

$$\frac{2.5}{10k} = \frac{V_{DS}}{10k} + \frac{4(V_{DS})^2}{10k}$$

$$2.5 - V_{DS} = 8((2)V_{DS} - \frac{V_{DS}^2}{2})$$

$$4V_{DS}^2 + V_{DS} - 2.5 = 0$$

$$0 = -4V_{DS}^2 + 17V_{DS} - 2.5$$

$V_{DS} = .675V$ or a neg V; choose pos

$V_{DS} = .153$ or 4.097 can't

$.153V$

$.675 + .5 = 1.175$

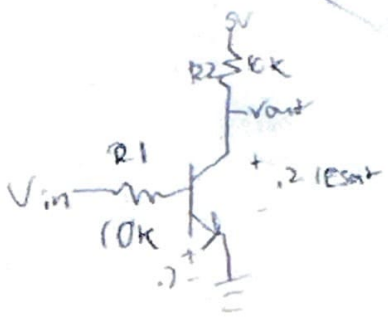
$.25V \rightarrow$ cutoff $= \boxed{2.5V}$

$.5V \rightarrow$ cutoff $= \boxed{2.5V}$

$1V \rightarrow$ sat $= \frac{2.5 - V_{DS}}{10k} = \frac{1}{2}\,40uA \cdot \frac{1}{2}(20)(1-.5)^2 \rightarrow 2.5 - V_{DS} = 1$ $\boxed{V_{DS} = 1.5V}$

$2.5 \rightarrow$ linear $= \boxed{.153V}$

2.



$\beta = 100$

assume active

$0 + 10kI_B + .7 = 0$ $I_C = 100(-20u) = -2mA$ $I_E = -2.02mA$

$I_B = -20uA$

$V_E = 0$

$\frac{0 - V_B}{10k} = -20uA$ ✓

$V_B = .7$

$.7 - .7 = 0$

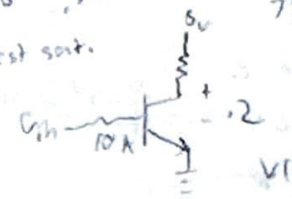
$\frac{5 - V_C}{10k} = -2mA$

$75 = V_C$??

No

$\frac{0 - 2.75}{10k} = R1_{curr}$

test sat.



$V_C = .2V$

$\frac{5 - .2}{10k} = 480uA$

$\frac{0 - .7}{10k} = -20uA$

Vin	Vout	R1 curr.	R2 curr	trans mode	Possible app.
0	.2V	-20uA	480uA	Sat.	load regulator
2	.2V	130uA	480uA	Sat	
3	.2V	230uA	480uA	Sat	
5	.2V	430uA	480uA	Sat	

2(a).

Assume Active

$-2 + 10k I_B + .7 = 0$

$10k I_B = 1.3$

$I_B = 130uA$

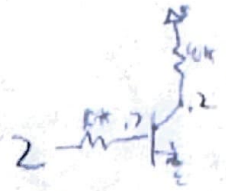
$I_E = 101 I_B = 13.13mA$

$I_C = 100 I_B = 13mA$

$V_E = 0$

$\frac{2 - V_B}{10k} = 130uA = .7 = .7 + V_E$ ✓

Sat



$\frac{5 - .2}{10k} = 480uA$

$\frac{2 - .7}{10k} = 130uA$

$\frac{5 - V_C}{10k} = 13mA$ - $V_C = 125$ Sat-saturation

assume active

$-3 + 10k I_B + .7 = 0$

$I_B = 230uA$

$\frac{3 - V_B}{10k} = 230uA$ $V_B = .7$ $V_E = 0$

$I_C = 100 I_B = 23mA$

$V_C = \frac{5 - V_C}{10k} = 23mA$ $V_C = -225$ Sat again

$\frac{3 - .7}{10k} = 230uA$ $\frac{5 - .2}{10k} = 480uA$

assume active

$-5 + 10k I_B + .7 = 0$

$4.3 = 10k I_B$ $I_B = 430uA$

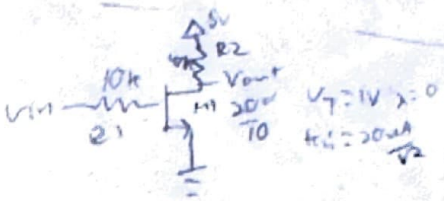
$\frac{5 - V_C}{10k} = 43mA$ $V_{out} = .2$

$V_C = -425$

Sat $\frac{5 - .2}{10k} = 480uA$

$\frac{5 - .7}{10k} = 430uA$

2b.



Vin	Vout	R1 curr	R2 curr	Mode of op.	Possible application
0	5V	0A	0A	cutoff	open switch
2	3V	0A	200uA	saturation	amplifier
3	.625V	0A	475uA	active	closed switch
5	.305V	0A	4695 uA	active	closed switch

$V_{in} = 0$ gate curr = 0

$V_g = 0$ $V_{GS} < V_T$ cutoff

$\frac{0 - V_g}{10k} = 0$

$V_{in} = 2$ assume sat

$V_B = 2$ 2>1

$V_{GS} = 2$

$\frac{5 - V_{out}}{10k} = \frac{1}{2} \frac{20uA}{V^2} \left(\frac{200}{10}\right)(2-1)^2$

$5 - 2 = V_{out}$ $V_{out} = 3$

$V_{out} = V_{DS}$

$3 > 2-1$ ✓ sat

2 ext.

$V_{in} = 3$ assume sat

$3 > 1$

$$\frac{5 - V_{out}}{10k} = \frac{1}{2}\frac{20mA}{V^2}(20)(2^2)$$

$5 - 8 = V_{out} = -3 \times$ not in sat

$$\frac{5 - .649}{10k}$$

active

$$\frac{5 - V_{DS}}{10k} = \frac{20mA}{V^2}(20)\left((3-1)V_{DS} - \frac{V_{DS}^2}{2}\right)$$

$$5 - V_{DS} = 4\left(2V_{DS} - \frac{V_{DS}^2}{2}\right)$$

$$5 = 9V_{DS} - 2V_{DS}^2$$

$$0 = -2V_{DS}^2 + 9V_{DS} - 5$$

$V_{DS} = .649$ or 3.851

b/c we're in active

so $V_{DS} < V_{GS} - 1$

$.649 < 3 - 1$ ✓

$V_{in} = 5$ assume active

$$\frac{5 - V_{DS}}{10k} = \frac{20mA}{V^2}(20)\left((5-1)V_{DS} - \frac{V_{DS}^2}{2}\right)$$

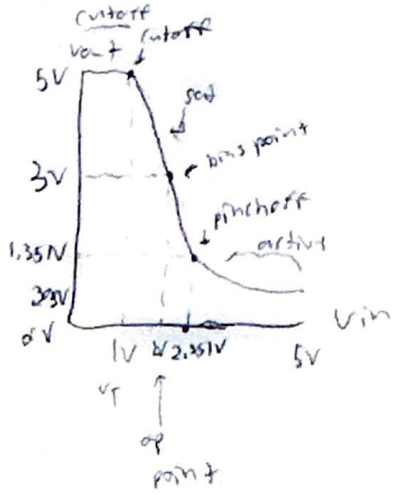
$$5 - V_{DS} = 4\left(4V_{DS} - \frac{V_{DS}^2}{2}\right)$$

$$0 = -2V_{DS}^2 + 17V_{DS} - 5$$

$.305$ or 8.195

$.305$ ✓

$$\frac{5 - .305}{10k}$$



$$\frac{V_{DD} - V_{DS}}{10k} = \frac{1}{2}k\frac{W}{L}(V_{DS})^2$$

$$\frac{5 - V_{DS}}{10k} = \frac{1}{2}\frac{20mA}{V^2}(20)(V_{DS})^2$$

$$5 - V_{DS} = 2(V_{DS})^2$$

$$0 = 2(V_{DS})^2 + V_{DS} - 5$$

$V_{DS} = 1.35$

$1.35 + 1 = 2.35$

d. gain mag = 4 V/V but actual gain will be −4 V/V

$g_m v_{gs} = -g_m R_D$

$-4 = -g_m 10k$

$g_m = 400 \frac{\mu A}{V}$

$400\mu A = 200\mu A/V^2 (2V)(V_{gs} - V_T)$

$1 = V_{gs} - 1$

$V_{gs} = 2$

$\frac{5 - V_{out}}{10k} = \frac{1}{2}\left(200\frac{\mu A}{V^2}\right)(2V)(2-1)^2$

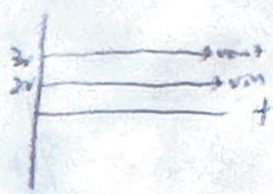
$5 - 2 = 3 = V_{out}$

It should be biased at 2V

Wouldn't minimal distortion just mean 0%? if so

$\frac{\hat{v}_{gs}}{4(V_{gs} - V_T)} \rightarrow \frac{\hat{v}_{gs}}{4} = 0$ so $\hat{v}_{gs} = 0$

peak input = 0



V_{out} is always $> V_{in} - V_T$

so the output will be undistorted